

20260624 Before Offline

Here I show the rMZR, $[\text{O}/\text{H}] - \Sigma_{\text{SFR}}$ relation and Spearman trends colorcoded with galaxy-level properties:

1. total H I mass ($\log M_{\text{HI}}$);
2. total H_2 mass ($\log M_{\text{H}_2}$);
3. total $M_{\text{HI}}/M_{\text{H}_2}$;
4. total $M_{\text{HI}}/(M_{\text{HI}} + M_{\text{H}_2})$;
5. total $M_{\text{H}_2}/(M_{\text{HI}} + M_{\text{H}_2})$.

All 3 ratio-binning plots show strong trends in upper (positive) and lower (negative) branches $[\text{O}/\text{H}] - \Sigma_{\text{SFR}}$ relation. So at least from the integrated point of view, we can see that the positive correlation is H_2 dominated while the negative trend is H_I dominated.











